# Player Skillset Grouping and Price Insights for Indian Premier League(IPL) Auction

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## ABSTRACT

In this work, we will be looking at player statistics from various sources and will be exploring individual players and differences/similarities between them. Also we have applied machine learning based algorithms that predicts the cost at which a player can be sold in the Indian Premier League Auction. In this we will use **Data Analysis Methods**: Principle Component Analysis, K-means Clustering, Chart & Plots. For **Auction Price Prediction Method**: Regression Trees, K-Nearest Neighbors(KNN), Linear Regression, Support Vector Regression, Lasso Regression, Ridge Regression

## KEYWORDS

IPL, Twenty20, Cricket, Machine Learning, Regression, Clustering

## 1 PROBLEM STATEMENT

The Indian Premier League (IPL) is a professional Twenty20 cricket league that was started in 2008 in India.[1] The IPL is the most-attended cricket league in the world and in 2019 was ranked third by average attendance among all sports leagues.[2] In 2019 the brand value of IPL was $6.3 Billion. [3]

The participating IPL franchises form their teams by conducting an auction before the tournament from a pool of national and international cricket players of different countries. As per IPL regulations, each franchise can spend a maximum of US $12.5m on their team salaries [4]. To select the correct player and predict the auction price of a batsman there are several important parameters to consider like "Batting average", "Batting strike rate","Consistency in Scoring Runs" etc. Similarly to predict the price of a bowler there are several important parameters to consider like "Economy rate", "Number of Wickets", etc. At present, the decision of choosing to bid for a player is done by eight representative from each franchises. Each member has different expertise and contributes to the final decision making. This makes it difficult to make decisions. Furthermore, pressure situations and hurried decisions which need to be made at the auction can lead to costly mistakes. The proposed model and data analysis makes the decision making more accurate, and well prepared for all possible scenarios that may arise. We are able to group players based on similar skill sets and categorize them according to their past records. We are also able to predict the price of a player based on his past performance. Our model helps the bidder realize that the exact range of the price the player is expected to be sold at, thereby enabling them to plan and adjust their budget. At present the accuracy with which auctioneers are able to evaluate a player is dependent on the experience in evaluating that the 8 members of the bidding team have. We aim to take the help of statistical analysis to eliminate some of the bias that the franchise members might bring to a table, and also give new insights that might not have been considered before.

## 2 DATA SOURCES

We are using player data of 2018–2021 IPL seasons from the following website iplt20.com and IPL Auction price data will be used from website cricmetric.com. We will also be leveraging data from the Indian Premier League (2008-2019) kaggle dataset which contains records of every ball bowled and every run scored from all IPL editions. This dataset will be used if we are finding difficulty in scraping data from iplt20.com, which contains a cleaner form of data that is useful. We would have to programmatically calculate the statistics from the kaggle dataset if we are finding difficulty with the iplt20.com stats.

## 3 METHODOLOGY

Players would be grouped based on different metrics such as strike rate, economy rate, number of wickets,

total runs scored, etc. The parent datasets would require manipulation to extract this information and group players so that franchises will be aware of players with skill sets. The above metrics will be used throughout our analysis of players. The steps to arrive at our grouping and price prediction insights are mentioned below:

**Solution Steps** -

We have divided the above problem statement into below workflow steps:

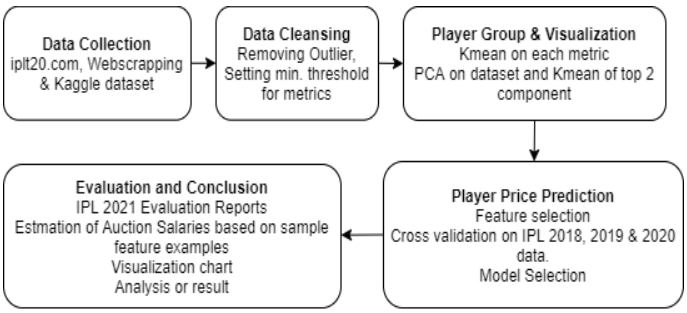



**Figure 1: Flow Diagram - Proposed Solution**

**Data Collection** The data would be scraped using the beautiful soup library from iplt20.com where we can find stats for batting and bowling. In the event that we have issues scraping the data, we can also obtain the same information programmatically using the kaggle dataset. The features that will be considered for the batsman are Highest Strike Rate, Most Number of Runs Scored and Best Batting Average. The features that will be considered for bowling are Wickets and Economy Rates. For price prediction, the additional features we would be using are Most 50s , Most Centuries, Most Fours, and Most Sixes. The additional features for bowling include Most Dot Balls, Most Bowling Average.

**Data Cleaning** Some basic cleaning has to be incorporated to ensure we have reliable data. For example, we may have to remove extraneous information that is not applicable that could potentially skew our results. If a batsman hit a six off of one ball, that will result in a huge strike rate. However, we will set a threshold of minimum number of balls to be played when taking strike rate into consideration. If we are utilizing the kaggle dataset, we need to remove extra columns and perform calculations to obtain the required metrics we need. We may have to also remove extraneous data here.

**Player Grouping** We will perform K-means on each individual feature such as strike rate, batting average, number of runs scored for batsman. For bowlers, I would perform K-means on economy rates and number of wickets. This gives us a detailed insight into grouping players into clusters based on each feature to identify players with similar skillsets. For overall grouping of players , I would also Perform PCA on all the features and take the top 2 components. I would perform Kmeans on the resultant components to group similar players on overall stats for both batting and bowling.

**Auction Price Prediction** Now that we have clusters of players available with similar skill sets, I would also provide insights into the likely auction price of each player, for budgeting purposes. I will consider the 2020,2019 and 2018 editions of the IPL and data from those editions for my training and validation. Cross-validation techniques will be used for training and validation. I would use the 2021 IPL statistics as my test data. The features would be the metrics that were mentioned previously. The independent variable would be the Auction salaries that would be obtained from another dataset.

The additional features would be added to the features used to group players and analysis is done to find the most significant features. The unnecessary features would be removed and different models such as classic regression, regression trees, SVM regression, Ridge Regression, and Lasso Regression will be evaluated. The model that has the best performance on the cross validation dataset will be used for predicting the 2021 season auction prices for each player. The accuracy of the model will be evaluated.

# 4 EVALUATION AND FINAL RESULTS

As mentioned previously there are 2 aspects to providing insights for the IPL Auction; grouping players based on similar skillsets and predicting auction prices. It is important that the models we train and grouping must be accurate as heavy deviations can be catastrophic and can lead to a huge financial waste.

**Evaluation of Grouping Players** The most important metric to consider validating the grouping of players are their salaries based on past IPL season data. Each grouping of players based on each feature should directly relate to a salary range ideally. It is safe to assume the best players are the most paid. There are exceptions

to this with new talent, but this is the best way to validate our groupings. If the maximum number of players within a cluster are within the same pay scale, then the grouping can be considered accurate.

**Evaluation of Player Auction Price Prediction** Each model that would be used in predicting the auction price of players would be evaluated on the basis of cross validation on the 2018,2019,2020 ipl seasons. The error in each model would be determined by the Auction Price of the Player-Model Predicted Auction Price of the Player. Once the best performing model is chosen, it is evaluated on the 2021 ipl dataset and player salaries are predicted. The error produced on the test dataset is a measure of evaluation of how well our model performed. If the error is within a reasonable threshold, then the model can be applied in future auctions given player metrics.

# REFERENCES


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